

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

DECEMBER EXAMINATION

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS
(EVENING AND WEEKEND CLASSES)

SATURDAY, 17 DECEMBER 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in Section A and **ONLY ONE (1)** question in Section B. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are /are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Section A (Compulsory Questions)**

Q1. (a) Given the table below:

Activity	Optimistic Time (Weeks)	Most Likely Time (Weeks)	Pessimistic Time (Weeks)	Predecessors
A	4	7	10	-
B	7	10	13	A
C	2	5	8	B
D	2	3	16	B
E	2	5	14	C
F	2	6	10	E
G	3	7	11	D, F

- (i) Calculate the expected time for the aforementioned activities. (7 marks)
- (ii) Draw a network diagram for the activities. (5 marks)
- (iii) Calculate and show the critical path. (5 marks)

- (b) Describe any **FOUR (4)** tasks to be completed by a project manager under **project initiation** phase. (8 marks)
- [Total : 25 marks]

Q2. (a) Discuss any **THREE (3)** types of acceptance testing performed before the delivery of a system to the users. (12 marks)

- (b) Company XU Enterprise is in the mid of installing a new system in their purchasing department. The management is a little skeptical on the performance of the new system thus decided to go for parallel installation.

- (i) Describe why **parallel installation** is suitable in this context. (6 marks)
- (ii) Illustrate any example of **parallel installation**. (4 marks)
- (iii) Highlight the main **disadvantage** of parallel installation. (3 marks)
- [Total : 25 marks]

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- Q3. (a) Maintenance referring to the changes made to a system to fix or to enhance its functionality. Corrective maintenance is one type of maintenance. Describe corrective maintenance. (10 marks)
- (b) Assuming monetary benefits of an information system at RM 95,000 per year, one-time costs of RM 81,000, recurring costs of RM 37,000 per year, a discount rate of 10%, and a three-year time horizon.
- (i) Calculate the net present value of these costs and benefits of an information systems. (6 marks)
- (ii) Calculate the overall return on investment of the project. (4 marks)
- (iii) Present a break-even analysis. (5 marks)
- [Note: Combine the answers of (i), (ii) and (iii) into one table]
- [Total : 25 marks]

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Section B (Choose any ONE Question)**

- Q4. (a) You are hired by Kensington Hospital to assist them in designing a hotel management system. The mentioned system should be able to record patients' data in order to generate related reports for the hospital. Only hospital staffs are allowed to handle patients' records. The admin gives access rights to the staff to handle the records in the system based on the list of selected staff given by the chief doctor. Apart from that, the system should be able to generate discharge bill to the patient based on their treatment. Upon receiving the payment from the patient, system should be able to update the data into patient record. Based on the given scenario, you are required to design the process modeling to assist further understanding.
- (i) Identify the sources and sinks. (6 marks)
 - (ii) List data flow between sources, sinks and main process. (6 marks)
 - (iii) Draw Context-Level DFD diagram. (5 marks)
- [NOTE: Combine the answer (i), (ii) and (iii) into a DFD diagram]
- (b) Joint Application Design (JAD) is an intensive group-oriented requirements technique in requirement determination phase.
- (i) Describe Joint Application Design (JAD). (4 marks)
 - (ii) List any **FOUR (4)** types of participants in Joint Application Design (JAD) session. (4 marks)
- [Total : 25 marks]

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Q5. (a) You are hired by Kensington Hospital to assist them in designing a hotel management system. The mentioned system should be able to classify a new case during registration at the reception. A nurse assesses a newly registered patient and enters the data into the system. The system then classifies them accordingly. If a patient is experiencing immediate life-threatening condition, then they are definitely at critical stage. However, the patient is only categorized as emergency if the patient has potential life-threatening situation and also showing some unstable vital signs. If there is no unstable vital sign, then they are just urgent case. The rest of the cases which does not meet the above conditions will be classified as non-urgent. Based on the given scenario, you are required to design the logic modeling to assist further understanding.

(i) List the conditions. (3 marks)

(ii) List the actions. (4 marks)

(iii) Prepare a decision table listing all possible rules. (10 marks)

[NOTE: Combine the answer (i), (ii) and (iii) into a decision table]

(b) Report is a business document that contains only predefined data. It is a passive document used solely for reading or viewing data. Discuss any **FOUR** (4) types of reports. (8 marks)

[Total : 25 marks]
